

1) Learning Goals:

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The learning goals for this project are:

1. *Master the use of common single stage amplifiers and multistage loading calculations*
2. *Apply all of the techniques that we have studied up to this point to a single design*
3. *Experience trading off parameters in a large analog system*

2) Design Project 1

.....

Design an amplifier over the next 1.5 weeks which meets the following specifications: $\hat{r}_{in} > 20k\Omega$ $\hat{r}_{out} < 50\Omega$ $\hat{a}_v = 1000 \pm 5\%$ $\hat{V}_{SW} > 5V$. $\hat{THD} < 5\%$ when $V_{sw} = 5V$.
Use only one 10 V power supply. (This constraint does not apply to your attenuator.) $\hat{}$

Table 1: Analytical, Simulated, and Measured Parameter Values for Amplifier

Parameter	Analytical	Simulated	Measured
Gain	1049.17	1010	1050
Input Resistance	21.5 k Ω	20.8 k Ω	20.8 k Ω
Output Resistance	17.95 Ω	17.86 Ω	22.3 Ω
Output Voltage Swing	9.17 V	8.5 V	8.8 V
THD at 5V Output Swing	N/A	0.17%	1.183%
Low Frequency Corner	N/A	2 kHz	2.5 kHz
High Frequency Corner	N/A	800 kHz	800 kHz
Power Consumption	48.2 mW	50.5 mW	52 mW
Stage 1 Collector Current	0.9803 mA	0.959 mA	0.961 mA
Stage 2 Collector Current	1.342 mA	1.28 mA	1.283 mA
Stage 3 Collector Current	1.0499 mA	0.112 mA	0.9616 mA
Stage 4 Collector Current	1.45 mA	1.55 mA	1.421 mA

Explanation of Design Process:

Given the performance specifications for the amplifier, we decided to bookend the stages with emitter followers. These configurations provide minimal output resistances relative to the other model types, making it desirable as the last stage, and significant input resistances, making it capable of satisfying the R_{in} constraint. This relatively high R_{in} , also helps mitigate the effects of interstage loading on the final stage, and the relatively low R_{out} , decreases interstage loading effects when transitioning to the second stage. In retrospect, these stages could have leveraged the 2N3904 NPN transistors to optimize performance relative to their β values; however, we opted to use the PNP transistors for these stages for no other reason than consistency.

A key concern when designing the intermediate stages was the effect of interstage loading on the overall gain. The middle two stages were chosen to be some form of common emitters and drove the overall gain of the system as the emitter follower stages only provide a gain of about 1. We initially had the second stage be the common emitter with degeneration and the third as a standard common emitter; however subsequent design optimization and interstage loading comparison indicated that swapping their positions was more ideal and flexible. To minimize power consumption, the collectors were targeted to run with a 1 mA current through them. Using the relationship that the current through the collector is about equal to the current through the emitter, we could tackle more aspects of the overall design. Resistors were the most fluid and interchangeable factor. Selecting reasonably-sized emitter resistors could help us determine which voltage should be at the top of the node. Using a V_{BEON} of 0.7 for the NPN transistors and understanding which voltage should be at the top of the emitter node, we could find the base voltage and the general relationship between the biasing resistors. With a known beta and collector current, we could use the general relationship of $\beta I_B = I_C$ to find out how big we should make these resistors. Making the resistors too small would mean that the current stolen by the base was negligible and reduce the droopiness of the voltage, but it would increase the power consumption. Making the resistors too large would reduce current, increasing the proportion of current stolen by the base, and impact the output voltage. Generally, we tried to stick to keeping the biasing network below 100 k Ω , but the last stage required us to modify this constraint to make sure that our analytical, simulation, and measured parameters were all simultaneously within specifications.

To make the gain more achievable an attenuating circuit was used to decrease a 100 mV input signal to 1 mV, prior to the amplifier. The attenuator included two resistor divider networks, with the second divider experiencing some loading from the first. This network fed into a dual-rail op-amp in unity gain to reduce signal noise.

To reduce interstage loading, the input resistance for each subsequent stage aimed to be at least one order of magnitude higher than the output resistance from the previous. The analytical approximations used to determine analytical performance and drive the design are present in Figure 1. The r_{in} values were modified to account for the biasing network, and the r_o resistor was assumed to be significantly large with minimal impact on performance. The design is fully realized in Figure 2.

The design process proceeded by first designing the analytical aspects, using equations, relationships, and intuition to understand the general layout and components necessary to meet the performance specifications. The resistor choices at the biasing point, the collector, and the emitter were the primary vehicles for modifying amplifier designs. Once all specifications were met analytically, we transitioned to the simulation phase where we conducted DC simulations to verify our bias points, AC simulations to verify our amplifier gain, and then transient simulations to verify the shape and voltage swing of the circuit. We tested each stage individually to isolate any problematic parts, then tested the entire amplifier as a whole to make sure the overall specifications were met. After these simulations, we used the transient behavior to validate R_{in} , R_{out} , and the Total Harmonic Distortion. After the simulations were all within specifications, we mirrored that process with the physical implementations and measurements. Any necessary design tweaks revealed in later stages of the design, prompted a reset, returning the the analytical components and climbing our way back again.

Amp	rin	rou	av
CE	r_π	$r_o R_C \approx R_C$	$-g_m R_C$
CE w/ degen (neglects r_o)	$r_\pi + (\beta + 1)(R_E r_o) \approx \beta R_E$	R_C	$\frac{-\beta R_C}{r_\pi + (\beta + 1)(R_E r_o)} \approx -\frac{R_C}{R_E}$
EF	$r_\pi + (\beta + 1)(R_E r_o) \approx \beta R_E$	$r_o R_E (1/g_m + R_S/\beta) \approx 1/g_m$	$\frac{(\beta + 1)(R_E r_o)}{r_\pi + (\beta + 1)(R_E r_o)} \approx 1$
CB	$\approx \frac{1}{g_m} + \frac{R_C}{g_m r_o}$	$\approx R_C \beta r_o$	$g_m R_C$
Cascode	$r_{\pi 1}$	$R_C (r_{\pi 1} r_{o1} + r_{o2} + \beta_2 r_{o2}) \approx R_C$	$-g_m R_C$

[illegible]

First the Circuit Was Designed on Paper, through the design process described above and as depicted by figure 2 below.

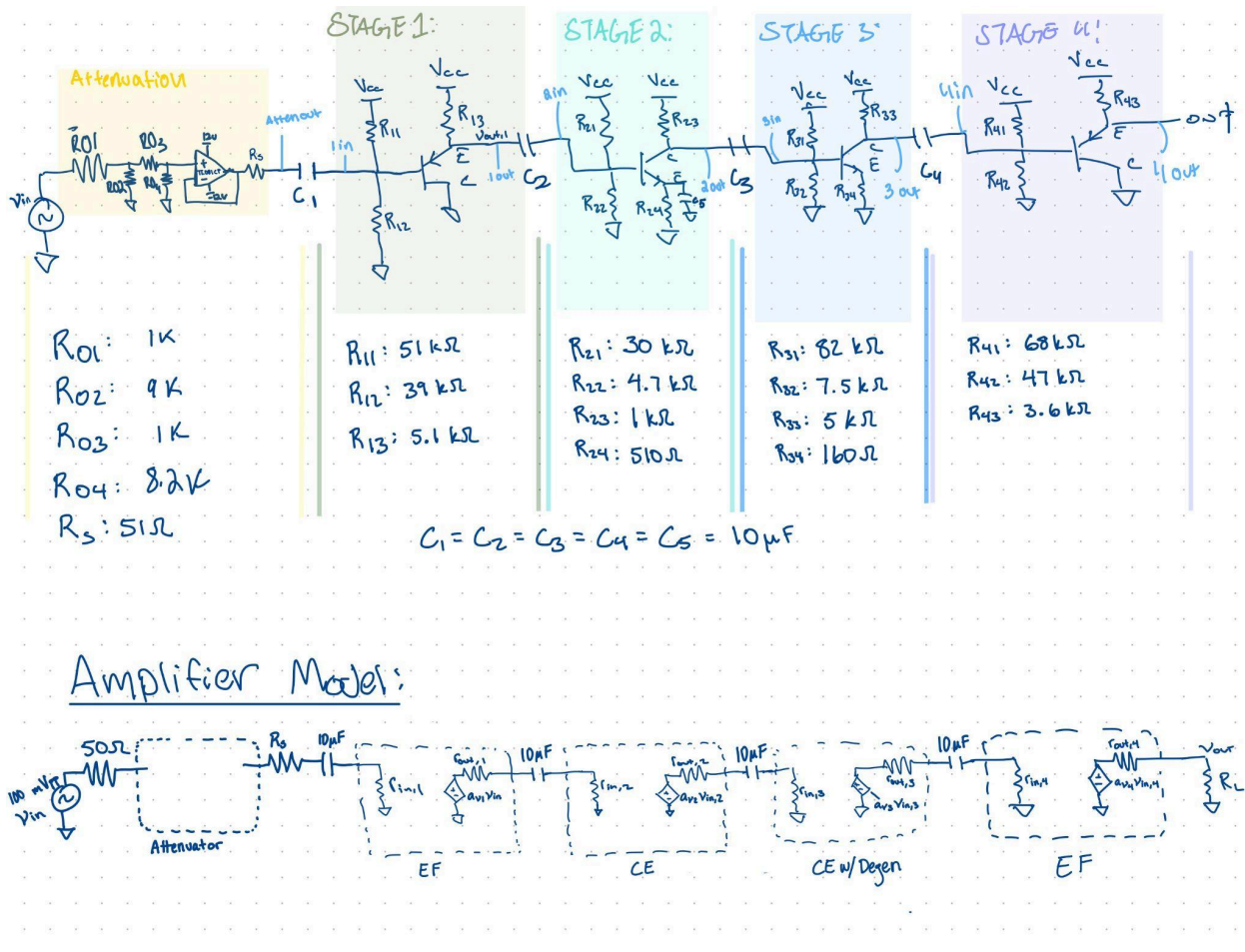
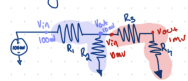


Figure1. Schematic of Circuit with Amplifier model shown below.

Attenuator Calcs:

Objective attenuate 100mV to 1mV.

Design a voltage dividers with 10x decimation



Voltage Divider 1:

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$$

$$10mV = \frac{R_2}{R_1 + R_2} (100mV)$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

$$\frac{1}{10} = \frac{R_2}{R_1 + R_2}$$

Voltage Divider 2

$$V_{out} = \frac{R_4}{R_3 + R_4} V_{in}$$

$$1mV = \frac{R_4}{R_3 + R_4} (10mV)$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

$$\frac{1}{10} = \frac{R_4}{R_3 + R_4}$$

Thus $R_1 = 1K\Omega$; $R_2 = 9K\Omega$; $R_3 = 1K\Omega$; $R_4 = 8.2K\Omega$

The attenuator was designed with two voltage dividers to convert 100mV into 1mV. After the two voltage divider stages a unity gain buffer was created with an op-amp in order to isolate the circuits and prevent loading. The second stage of the divider accounts for loading from the first.

Figure 2: Attenuator Calculations. The attenuator was designed with two voltage dividers to convert 100mV into 1mV.

Next the circuit was built and simulated in LT Spice Using sub circuits assembled into the attenuator and amplifier as depicted in figure 3 below.

Screenshot of LTSpice schematic

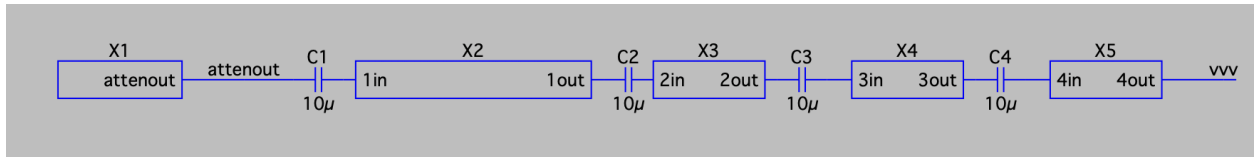


Figure 3. Screenshot of LTSpice Schematic

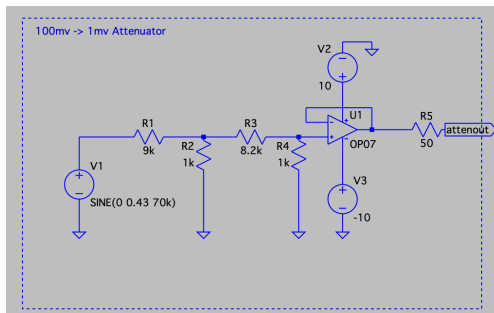


Figure 3a.. Attenuator LTSpice Schematic

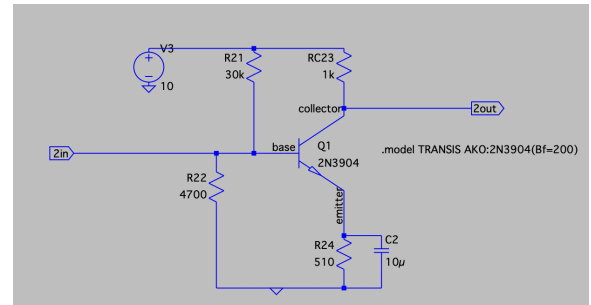


Figure 3b. Stage 1 LTSpice Schematic

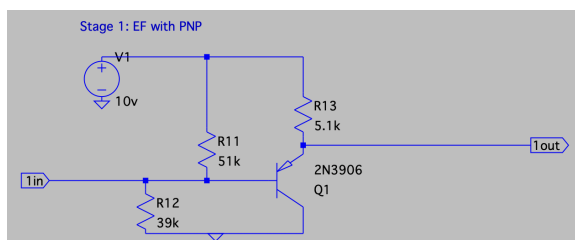


Figure 3c.. Stage 2 LTSpice Schematic

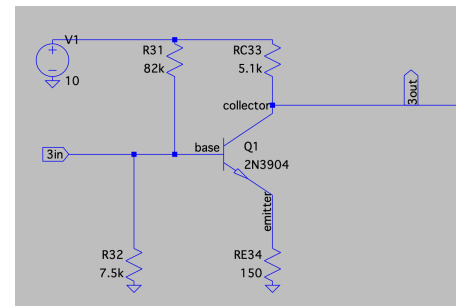


Figure 3d. Stage 3 LTSpice Schematic

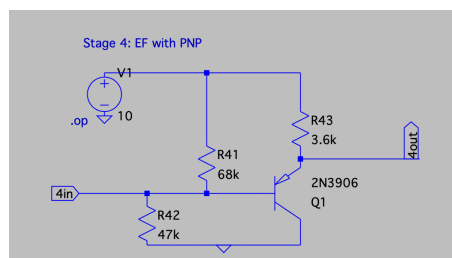


Figure 3e. Stage 4 LTSpice Schematic

Finally the Circuit was assembled on a breadboard to test a real version of the design:

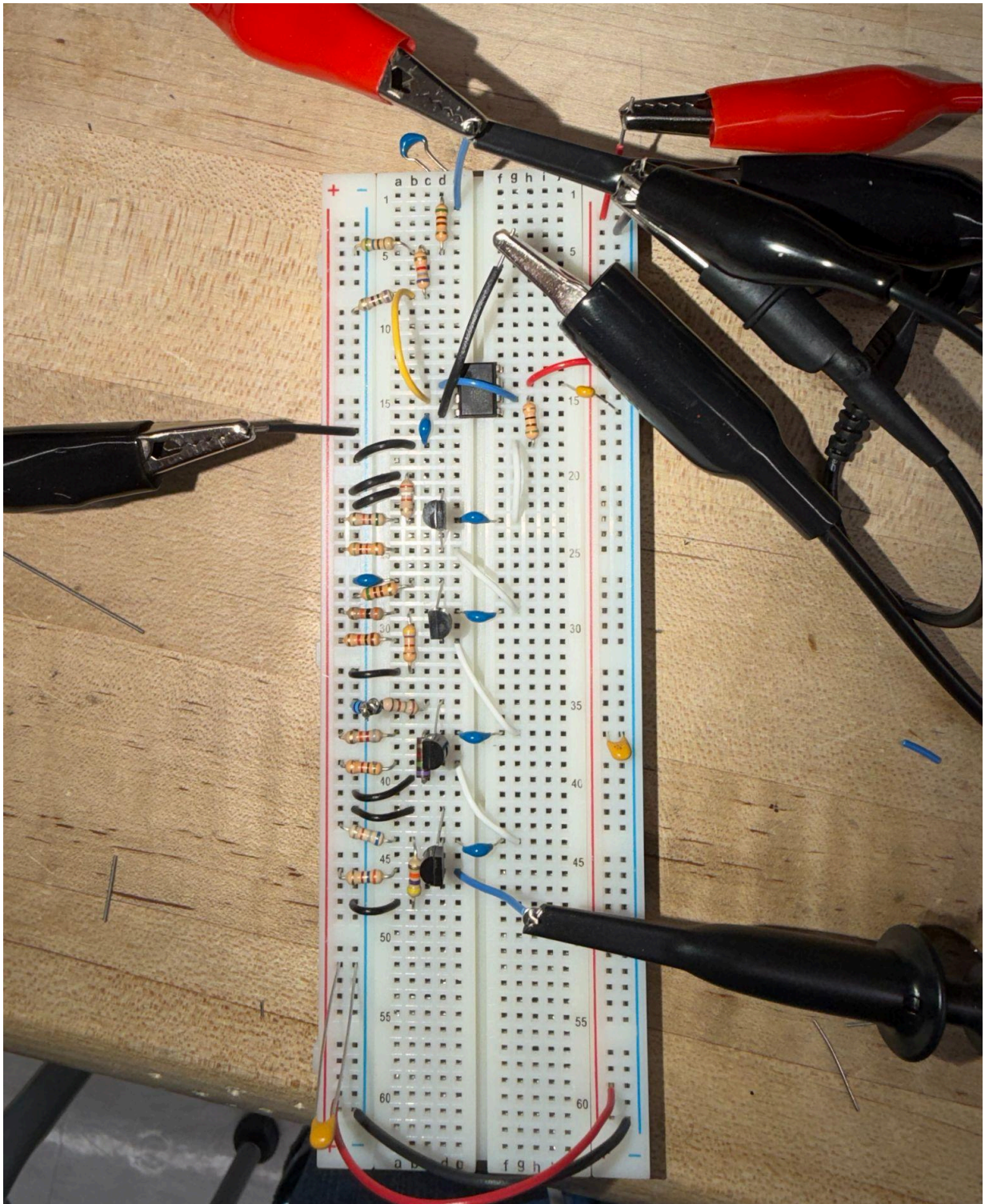


Figure 4. Picture of Circuit in Breadboard

When tested the real circuit showed normal amplifying operation through its input and output traces

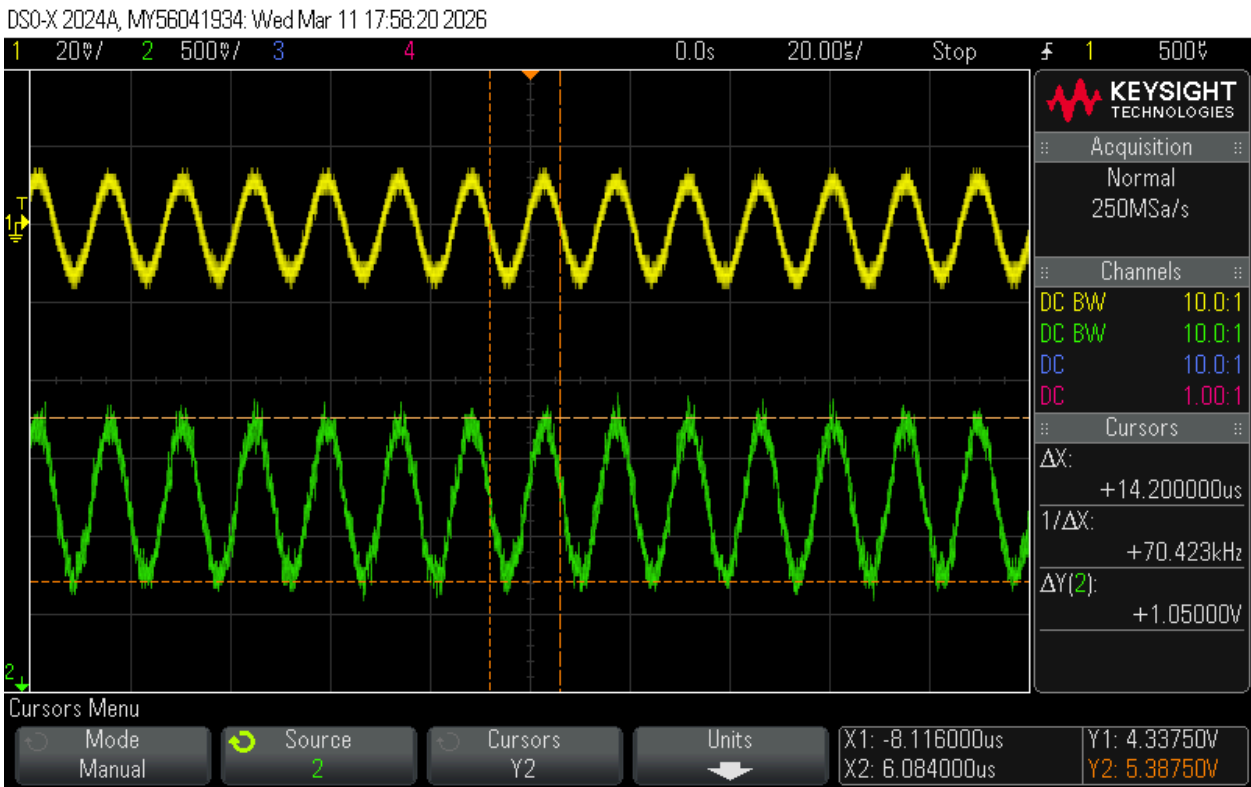


Figure 5. Input and output showing normal amplifying operation with a gain of 1,050. Yellow represents input, green represents output. Input and output traces further showed maximum output voltage swing of 8.8 V (Figure 5) that meet THD spec (Figure 6).

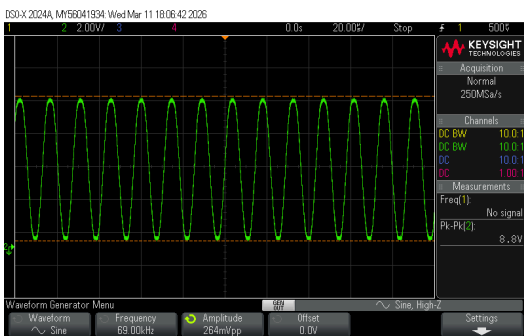


Figure 5. Voltage Swing found by raising input amplitude until the sinusoid clipped at 8.8 V PP which indicates a voltage swing of 8.8V

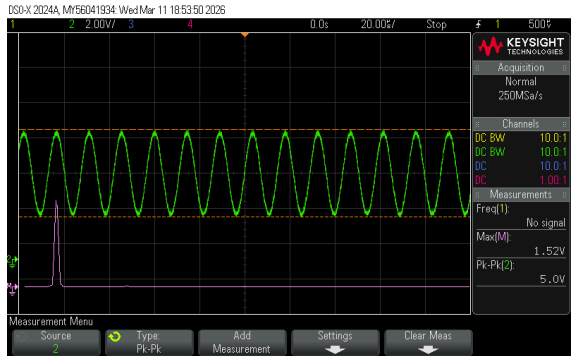


Figure 6. FFT of maximum output swing showing THD of 1.1%

The measured voltage swing, FFT and THD matched the simulated voltage swing, FFT and THD which are depicted in the figures 7-9 below. There is no schematic for voltage swing set up as it was the same as our main setup (Figure 2)

Voltage swing is defined by the max peak to peak voltage before clipping occurs. In order to find the voltage swing input voltage was increased to approximately 500mV in both measured Vsw and the simulated VsW.

$$\text{Equation 1} \quad \text{Voltage Swing} = \text{Upper limit} - \text{Lower Limit}$$

Voltage swing was simulated to be 8.8 VPP and Measured to be 8.8 VPP- both matched as expected .

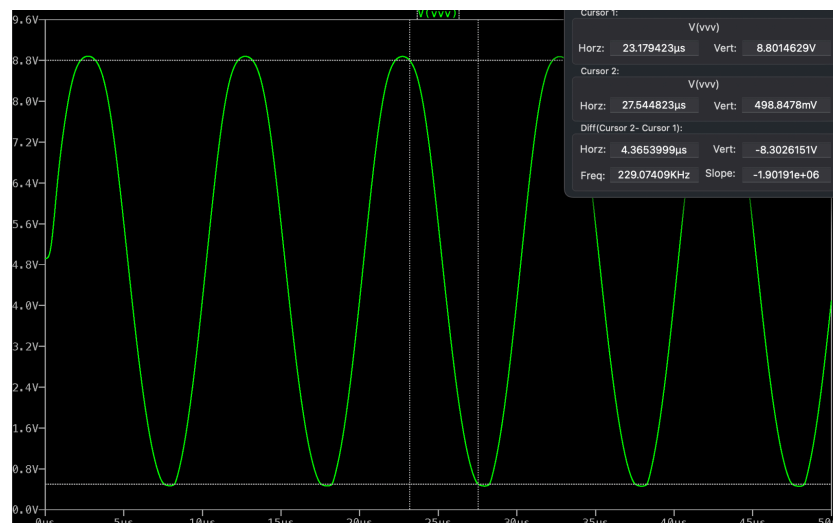


Figure 7. Simulated Voltage Swing at 8.8 V pp

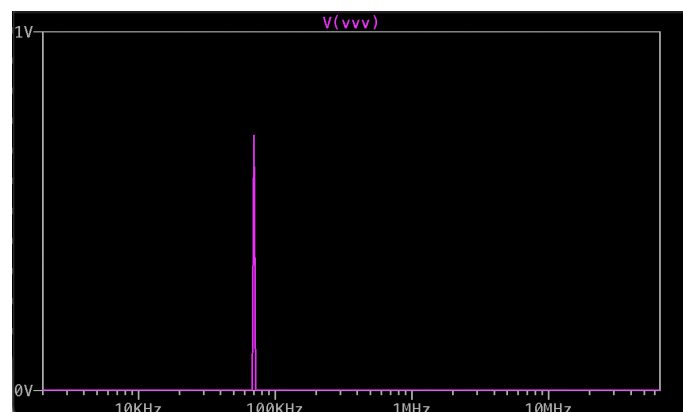
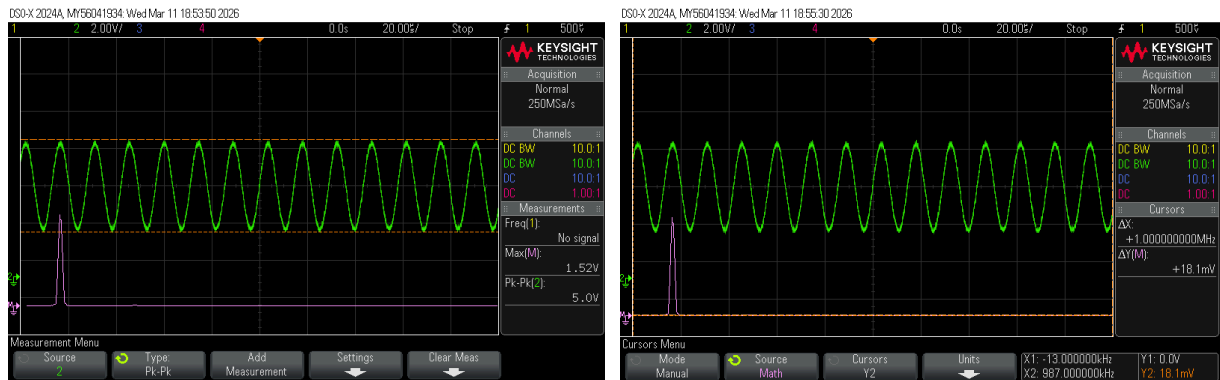


Figure 8. Simulated FFT

Partial Harmonic Distortion: 0.154313%
Total Harmonic Distortion: 0.174715%

Figure 9. Simulation of Maximum Output Swing FFT THD of .17%



Measured total harmonic distortion measured at a 5 V voltage swing. Only one minor harmonic was present with the most significant contribution from the fundamental frequency. Scope cursors were used to track the height of each peak and the equation below was used to calculate the distortion:

Equation 2

$$THD = \sqrt{\frac{V_2^2 + V_3^2 + \dots}{V_1^2}}$$

Plugging in $V_1 = 1.53 \text{ V}$; $V_2 = .0181 \text{ V}$ and no other perceptible harmonics we get a measured THD of 1.183% well within spec and close to our simulated voltage swing. The increase in THD between simulated and measured FFT resonance peaks is explained by the increase in noise and introduction of nonidealities when dealing with real components compared to idealized simulations

BODE PLOTS

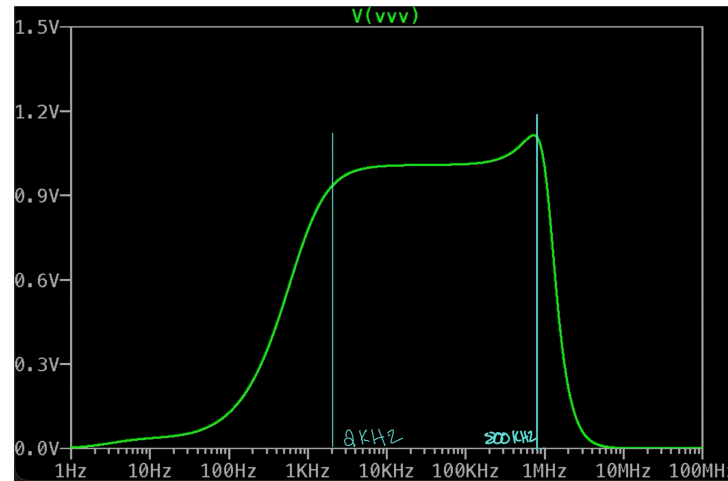


Figure 10. Simulated Bode Plot with band gap and 1.010 V showing desired gain a_v of 1,010 and f_{high} at 800 kHz and f_{low} at 2 kHz.

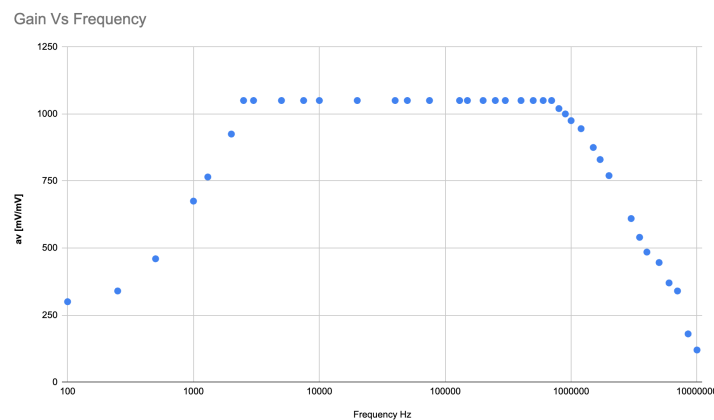


Figure 11. Measured Bode Plot with Band gap at 1.050 V showing desired gain a_v of 1,050 with f_{high} at 800 kHz and f_{low} at 2.5 kHz.

To generate the Bode plot (Figure 11), we applied a 1 mVpp sinusoidal input signal and swept the input frequency across the operating range while measuring the corresponding output voltage. This procedure allowed us to determine the amplifier's low and high cutoff frequencies as well as the mid-band gain. The Measured f_{low} and f_{high} values were very similar- these values were determined by looking at the frequencies at which the bode plots had their corners. The band gaps were also similar close to the desired gain of 1000 and within 10% of one another.

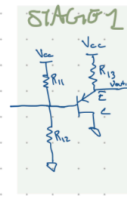
Notably, the resonant peak predicted in simulation was not observed in the measured frequency response. This discrepancy could potentially be attributed to parasitic capacitances or component tolerances in the physical circuit. Implementing additional filtering stages may help achieve the simulated resonant behavior in future iterations. The noise of the physical measurements and scope resolution introduced an added degree of human error that could influence readings.

R_{in}R_{in} analytical:

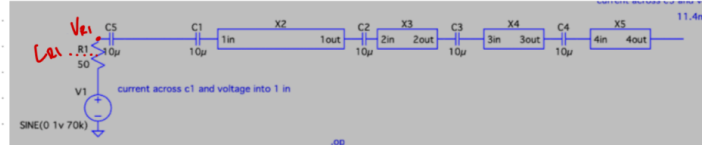
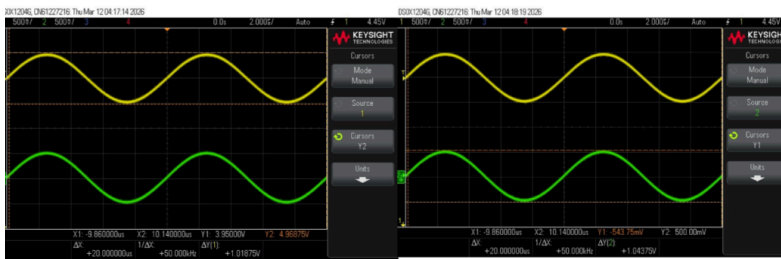
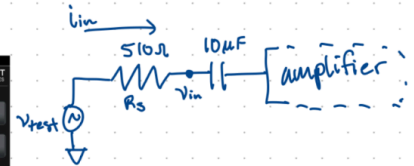
$$r_{in} = \left(\frac{1}{\beta \cdot R_E} + \frac{1}{R_{11}} + \frac{1}{R_{12}} \right)^{-1}$$

$$r_{in} = \left(\frac{1}{150 \cdot 800\Omega} + \frac{1}{51k\Omega} + \frac{1}{39k\Omega} \right)^{-1}$$

$$r_{in} = 21.479k\Omega$$

R_{in} SimulatedR_{in} calc:

$$r_{in} = \frac{V_{in}}{Current\ R_{in}} = \frac{1V}{48\mu A} = 20.8k\Omega$$

Test bench:R_{in} MeasuredYellow: $V_{in} = 1.0875V$ Green: $V_{test} = 1.04375V$ 

$$I_{in} = \frac{V_{test} - V_{in}}{R_s} = 4.902 \times 10^{-5} A$$

$$r_{in} = \frac{V_{in}}{I_{in}} = 20762.5\Omega$$

Figure 12: Analytical, Simulated, and Measured R_{in} Calculations

All measurements and calculations for the amplifier's input resistance were within 10% of each other, indicating high congruence between each modeling method. The analytical calculations demonstrate the highest estimation for R_{in} as mathematical approximations were used rather than precise equations. Some resistors like r_o were omitted from the calculations, although in reality they provide some contribution to overall r_{in} measurements for emitter followers.

The simulation likely provides the most accurate theoretical approximation. The R_{in} value was found by using a test voltage circuit and omitting the attenuation stage that precedes the amplifier. A sense resistor, coupling capacitor, and 1 V_{pp} input were required to carry out this stage. The simulation allowed for the input current to be probed and the input voltage was found at the exit terminal of the resistor. The input resistance was found by taking V_{in}/I_{in} .

A similar approach to the simulation was used for the physical measurements, as the same test circuit and testing conditions were used (as demonstrated in the circuit below).

Resistor tolerance, breadboard capacitance, noise, and oscilloscope measurement were primary roadblocks in finding the input resistance. This prompted frequent redesigns, increasing the sense resistor by an order of magnitude, and careful analysis of the scope probes. Unlike most other measurements, the inherent peak-to-peak voltage measurement that oscilloscopes provide could not produce the necessary resolution, so cursor measurements were required. R_{in} was calculated in the same manner by dividing v_{in} by i_{in} . The input current was found by finding the voltage drop across the resistor, and dividing it by the resistance value. The input voltage was the voltage measured at the exit of the resistor before entering the amplifier.

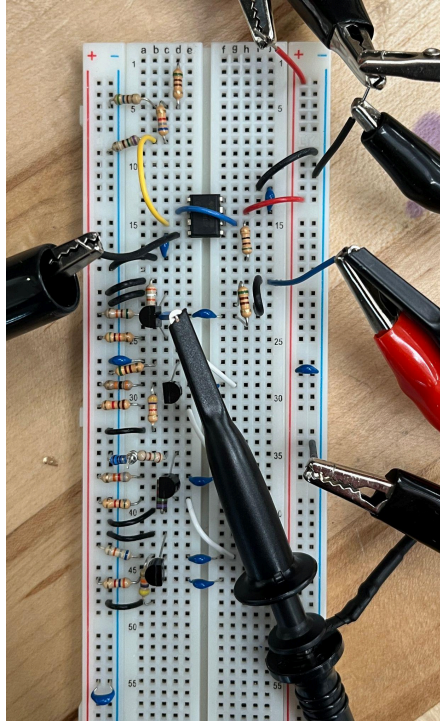


Figure 13: Circuit Setup for R_{in} Measurement

Analytical calculation for r_{out} , schematic of r_{out} test setup, brief explanation of test, calculations showing how to get from data to r_{out} . Explanations of discrepancies between analysis, simulation and measurement. Reference earlier schematics or results and use equations as necessary. (1 page)

R_{out}:

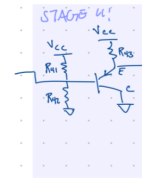
The analytical output resistance for the amplifier was found by using the emitter follower output resistance approximation, $1/g_m$. (Figure 14) Similar to the R_{in} calculations, the more precise equations could yield more accurate results; however, relative to the simulation, the estimated R_{out} appears to be satisfactory and within 10%.

A test circuit (Figure 15) was applied at the output of the amplifier, with the circuit drawn below. Similar specifications for the simulated R_{in} measurements were used here. A 1 Volt peak-to-peak input was used with a $50\ \Omega$ sense resistor and coupling capacitor. The attenuator was omitted and the original AC input node in stage one was grounded. All values in the simulation are considered ideal and measurements could be made by scoping the currents and voltages at the input nodes. Some human error may be present with reading and tracing the graphs; however, we expect these to be the most theoretically accurate.

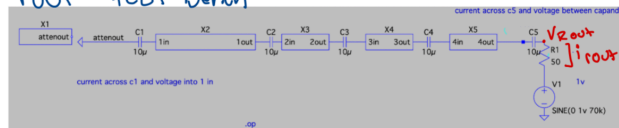
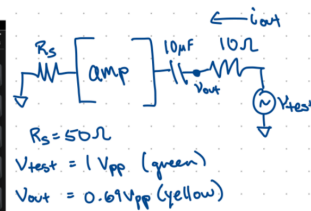
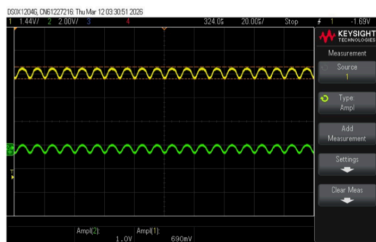
The measured output resistance (figure 14) demonstrates the most significant deviance from the analytical and simulated results. With this test in particular, we encountered significant issues with noise, breadboard capacitance, and oscilloscope reliability. The measured voltages were often many milliVolts off from what their input values were and slight oscillations in the 10 V power supply rails showed frequent oscillations. The presence of our bodies nearby also seemed to have an effect in signal behavior. We cannot confidently state that the measured R_{out} is the most accurate result; however, the scope traces were collected when the signal appeared to settle. The source resistance prior to the input grounding may cause deviances from the simulated and actual measurements as well. The sense resistor at the output was decreased by a factor of 5 as well due to the oscilloscope's sensitivity and resolution. Similar to R_{in} , the measured R_{out} was found by finding the voltage drop across the sense resistor and dividing that by the resistor value to get the input current. The voltage at the node prior to the coupling capacitor at the output represented the voltage out node. That voltage was divided by the output current to get R_{out} .

R out continued:R_{out} analytical:

$$r_{out} = \frac{1}{g_m} = \frac{i_c}{\phi_m} = \frac{.00145}{.026} = 17.95 \Omega$$

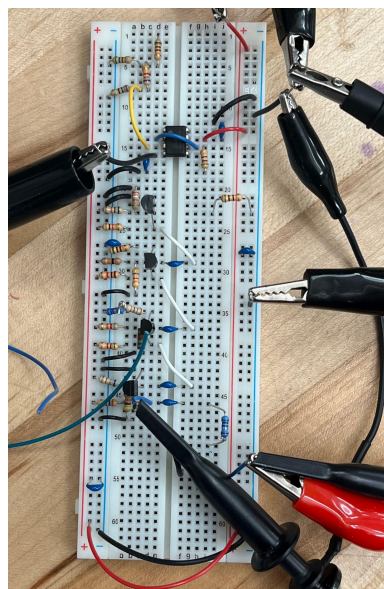
R_{out} Simulated

$$r_{out} = \frac{v_{out}}{i_{out}} = \frac{410 \text{ mV}}{23 \text{ mA}} = 17.86$$

R_{out} test benchR_{out} measured

$$i_{out} = \frac{V_{test} - V_{out}}{10 \Omega} = 0.031 \text{ A}$$

$$r_{out} = \frac{V_{out}}{i_{out}} = 22.3 \Omega$$

Figure 14: Analytical, Simulated, and Measured R_{out} CalculationsFigure 15: Circuit setup for R_{out} calculations

av

Analytical a_v calculations:Stage 1 - PNP EF:

$$r_{in,1} \approx \beta R_E \parallel R_{B1} \parallel R_{B2} = 21479 \Omega \quad a_{v,1} = \frac{(\beta+1)(R_E \parallel r_o)}{r_{in,1} + (\beta+1)(R_E \parallel r_o)} = 0.995 \quad r_{out,1} = \frac{1}{g_m} = 26.9 \Omega$$

Stage 2 - NPN CE:

$$r_{in,2} = R_{B1} \parallel R_{B2} \parallel R_{E1} = 1983 \Omega \quad a_{v,2} = -g_m R_C = -51.6 \quad r_{out,2} = R_C \parallel r_o = R_C = 1000 \Omega$$

Stage 3 - NPN CE w/ Degn:

$$r_{in,3} = \beta R_E \parallel R_{B1} \parallel R_{B2} = 10609.5 \Omega \quad a_{v,3} = -\frac{R_C}{R_E} = -26.945 \quad r_{out,3} = R_C = 5000 \Omega$$

Stage 4 - PNP EF:

$$r_{in,4} = \beta R_E \parallel R_{B1} \parallel R_{B2} = 26431 \Omega \quad a_{v,4} = \frac{(\beta+1)(R_E \parallel r_o)}{r_{in,4} + (\beta+1)(R_E \parallel r_o)} \approx 1 \quad r_{out,4} = \frac{1}{g_m} = 17.95 \Omega$$

Total gain with interstage loading:

$$a_{v,total} = a_{v,1} \left(\frac{r_{in,2}}{r_{out,1} + r_{in,2}} \right) a_{v,2} \left(\frac{r_{in,3}}{r_{out,2} + r_{in,3}} \right) a_{v,3} \left(\frac{r_{in,4}}{r_{out,3} + r_{in,4}} \right) a_{v,4}$$

$$a_{v,total} = 1049.17$$

Figure 16: Analytical Calculations of Individual Stage and Full Amplifier Gains (with interstage loading) and Adjusted for Biasing

To calculate a_v analytically the a_v was calculated for each stage individually and scaled with interstage loading. The a_v formulas came from the types of emitters and the beta values of the types of transistors.

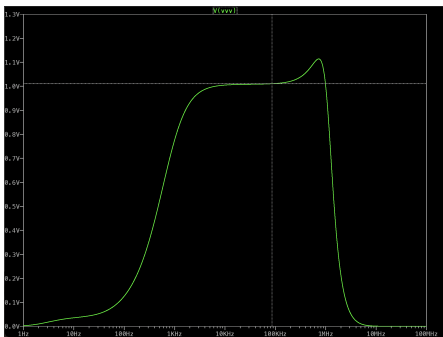


Figure 17. Simulated Bode Plot showing $a_v = 1,010$ gain

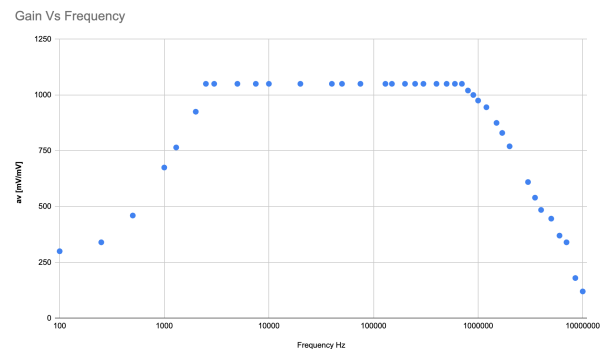


Figure 18. Measured Bode Plot showing $a_v = 1050$ gain

In Figure 16 a gain of 1049.17 was calculated analytically, this gain was reduced in simulation to 1010 (Figure 17). This discrepancy in gain is likely due to the non-idealities that are not taken into account in an analytical approach that are more included in a SPICE model. The discrepancy between simulated and measured are likely due to tolerances of components such as resistors that are often a small percentage off from their reported value. However the difference between the measured gain (Figure 18) and calculated gain (Figure 16) was less than 0.1% which means that the circuit performed as designed when under test.

VSW

Analytical voltage swing was found by taking the lowest value between the analytical voltage swing up and voltage swing down and multiplying that value by two.

Equation 3: $VSW = 2 \times \min (V_{cc} - V_{out}, V_{out} - V_{c_{stage\ 4}} - 0.2)$

Analytical Voltage Swing was found to be 9.17 V. This is within 10% of the simulated and experimental voltage swings. This indicates that the design was accurate and predictive of simulated and measured results. The slight reduction in voltage swing is likely due to nonidealities in simulation and experimental setups. Experimental voltage swing, 8.8 V, was found in figure 5 and simulated voltage swing, 8.5 V, was found in figure 7.

P

In order to calculate P (Power) the following equation was used:

$$\text{Equation 4} \quad P = V_{cc} (I_{c1} + I_{c2} + I_{c3} + I_{c4})$$

Analytically, P was calculated using the collector currents we found by dividing the output voltage by the resistor values. Then the sum of the collector currents were multiplied by Vcc of 10V which was 48.2m.

Table 2: Calculations for Test (left) and Simulated (right)

	Voltage Drop Test	Resistance	Currents	Voltage Drop Sim	Resistance	Current Sim
stage 4 emitter	5.116	3600	0.001421111111		3600	0.0014107977
r41	5.767	68000	0.000084808823		68000	0.000843139895
r33	4.808	5000	0.0009616		5000	0.000112
r31	9.14	82000	0.000111463414		82000	0.0001
r23	1.283	1000	0.001283		1000	0.00128
r21	8.63	30000	0.000287666666		30000	0.000228
r13	4.901	5100	0.000960980392		5100	0.000963299
r11	5.545	51000	0.000108725490		51000	0.000109
SUM		TOTAL CURRENT	0.005219355898		TOTAL CURRENT	0.005046236595
	vcc	POWER			Power	
	10	0.05219355898	52 mW		0.05046236595	50.5mW

In order to find the power of the circuit experimentally the measured voltage drops and known resistance values were used in accordance with ohms law the current of resistors connected to Vcc were found by dividing the voltage drop by the resistance. This same approach was used for the simulation, however for the simulation the currents could be measured directly. The currents were summed and then multiplied by Vcc. The Simulated power was found to be 50.5 mW (Table 3) and the measured power was found to be 52 mW (Table 3).

Table 3: Power Consumption Results

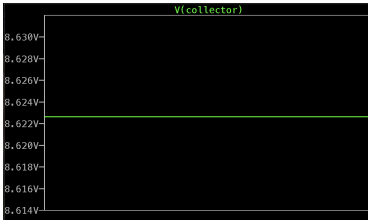
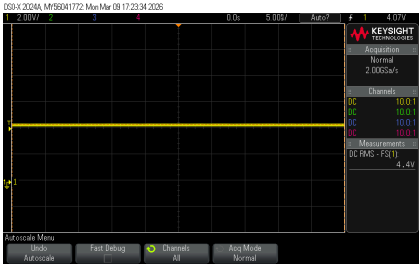
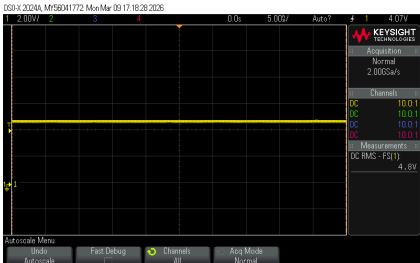
	Analytical	Simulated	Measured
Power Consumption	48.2 mW	50.5 mW	52 mW

The differences in power were small, this likely due to the nonidealities being better represented in simulation and more present in measured values. The 3 values were all within 10% of each other so it can be concluded that the analytical simulated and measured power consumption results were representative of one another as desired.

Flow and Fhigh were found earlier on page 10 through bode plots and the results are discussed on page 10.

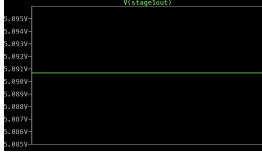
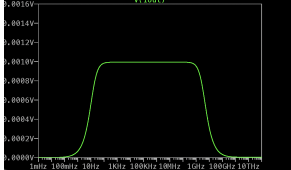
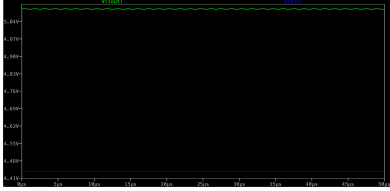
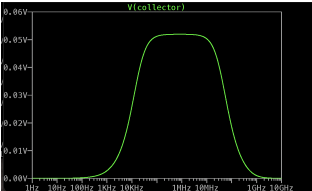
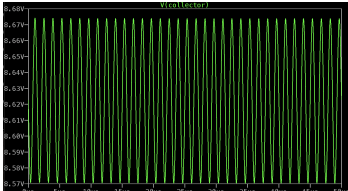
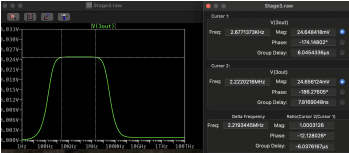
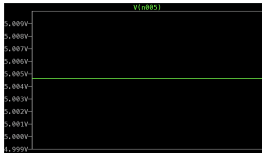
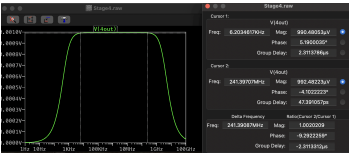
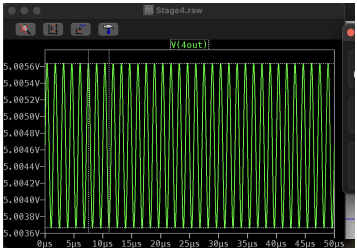
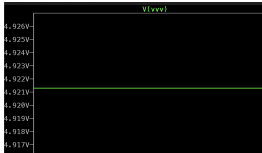
DC Results:

Table 4: DC operating points summary Table

	DC operating Points Calculated	DC operating points Simulated	DC operating Points Experimental
Stage1	5V	 5.09V	 5V
Stage2	8.65V	 8.623 V	 8.6V
Stage 3	4.96V	 4.99 V	 4.4V
Stage 4	4.79	 5V	 4.8V

Explanation of design process. Reference pages 2-3

APPENDIX A- Simulated results

	DC operating points	AC gain	Transient Output Voltage
Stage1	 DC 5.09V	i 	
Stage2	 DC 8.623		
Stage 3	 DC 6.068		
Stage 4	 DC 5V		
All stages together output	 4.9215	